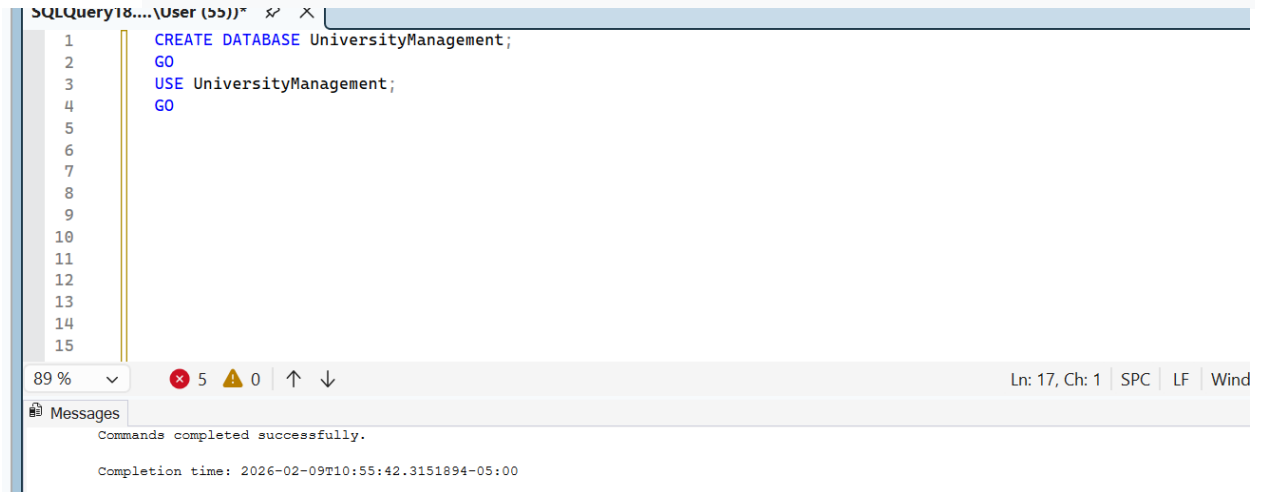


1. Database Creation:

- a. Create a database named UniversityManagement.



The screenshot displays the SQL Server Enterprise Manager interface. The top pane shows a query window titled 'SQLQuery18....\User (55))' containing the following T-SQL commands:

```
1 CREATE DATABASE UniversityManagement;  
2 GO  
3 USE UniversityManagement;  
4 GO  
5  
6  
7  
8  
9  
10  
11  
12  
13  
14  
15
```

The bottom pane shows the 'Messages' window with the following output:

```
Commands completed successfully.  
Completion time: 2026-02-09T10:55:42.3151894-05:00
```

The status bar at the bottom indicates 'Ln: 17, Ch: 1 | SPC | LF | Wind'.

2. Define Tables:

- Create all tables as per the entity definitions, with appropriate constraints:
 - Primary and foreign keys.
 - Unique, NOT NULL, and CHECK constraints where applicable.

```
o
9      --Departmental & Faculty Tables
.0  CREATE TABLE Departments (
.1      DepartmentID INT PRIMARY KEY,
.2      DepartmentName VARCHAR(100) NOT NULL UNIQUE,
.3      HeadOfDepartment VARCHAR(100)
.4  );
.5
.6
.7
.8
```

% 3 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99

Ln: 16, Ch: 1 | SPC

Messages

Commands completed successfully.

1.

Completion time: 2026-02-09T10:57:40.6055004-05:00

```
14  CREATE TABLE Faculty (
15      FacultyID INT PRIMARY KEY,
16      Name VARCHAR(100) NOT NULL,
17      Designation VARCHAR(50),
18      DepartmentID INT,
19      FOREIGN KEY (DepartmentID) REFERENCES Departments(DepartmentID)
20  );
21
22
23
24
25
26
```

% 3 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99

Ln: 23, Ch: 1 | S

Messages

Commands completed successfully.

Completion time: 2026-02-09T10:59:04.4788297-05:00

```
22  --Students and courses
23
24  CREATE TABLE Students (
25      StudentID INT PRIMARY KEY,
26      Name VARCHAR(100) NOT NULL,
27      DOB DATE,
28      DepartmentID INT,
29      FOREIGN KEY (DepartmentID) REFERENCES Departments(DepartmentID)
30  );
31
32
33
34
```

39 % 2 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99

Ln: 32, Ch: 1 | SPC | LF | Window

Messages

Commands completed successfully.

Completion time: 2026-02-09T11:00:04.9079979-05:00

```
32
33  CREATE TABLE Courses (
34      CourseID INT PRIMARY KEY,
35      CourseName VARCHAR(100) NOT NULL,
36      Credits INT CHECK (Credits > 0), -- Stored as number per hint
37      DepartmentID INT,
38      FacultyID INT,
39      FOREIGN KEY (DepartmentID) REFERENCES Departments(DepartmentID),
40      FOREIGN KEY (FacultyID) REFERENCES Faculty(FacultyID)
41  );
```

89 % 2 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99

Ln: 37, Ch: 22 |

Messages

Commands completed successfully.

Completion time: 2026-02-09T11:00:46.1765008-05:00

```
36
37 --Enrollment and Library
38
39 CREATE TABLE Enrollments (
40     EnrollmentID INT PRIMARY KEY,
41     StudentID INT NOT NULL,
42     CourseID INT NOT NULL,
43     Grade DECIMAL(3,2) CHECK (Grade >= 0 AND Grade <= 4.0), -- Stored as number
44     FOREIGN KEY (StudentID) REFERENCES Students(StudentID),
45     FOREIGN KEY (CourseID) REFERENCES Courses(CourseID)
46 );
47
48
49
89 % 1 0 1 0
Ln: 48, Ch: 1 SPC
Messages
Commands completed successfully.
Completion time: 2026-02-09T11:01:46.1580155-05:00
40
49 CREATE TABLE Library (
50     BookID INT PRIMARY KEY,
51     Title VARCHAR(255) NOT NULL,
52     Author VARCHAR(100),
53     BorrowerID INT, -- Links to StudentID
54     DueDate DATE,
55     FOREIGN KEY (BorrowerID) REFERENCES Students(StudentID)
56 );
57
58
59
% 1 0 1 0
Ln: 57, Ch: 1 SPC LF Window
Messages
Commands completed successfully.
Completion time: 2026-02-09T11:02:39.4359519-05:00
```

Indexes:

a. Create clustered and nonclustered indexes for query optimization.

```
57
58 --Optimization index
59
60 -- Non-clustered index for searching students by name
61 CREATE NONCLUSTERED INDEX IX_StudentName ON Students(Name);
62
63 -- Non-clustered index for checking enrollment by Course
64 CREATE NONCLUSTERED INDEX IX_Enrollment_Course ON Enrollments(CourseID);
65
66 -- Non-clustered index for finding books by DueDate
67 CREATE NONCLUSTERED INDEX IX_Library_DueDate ON Library(DueDate);
89 % No issues found
Ln: 67, Ch: 66 SPC LF
Messages
Commands completed successfully.
Completion time: 2026-02-09T11:05:12.2585856-05:00
```

Part 3: Populate the Database

1. Insert Data:

a. Populate each table with at least 10 records of realistic sample data.

- b. Ensure meaningful relationships (e.g., a student belongs to a department and is enrolled in courses offered by that department).

The screenshot displays the SQL Server Enterprise Manager interface. The top pane shows the execution of two SQL queries. The first query, SQLQuery18, populates the Departments and Faculty tables. The second query, SQLQuery19, selects the top 1000 courses from the UniversityManagement database. The bottom pane shows the results of the second query, displaying a table with columns CourseID, CourseName, Credits, DepartmentID, and FacultyID. The Object Explorer on the left shows the database structure, including the UniversityManagement database and its tables.

```
--POPULATING DATA

-- Populating Departments (Modern Tech & Science Leaders)
INSERT INTO Departments (DepartmentID, DepartmentName, HeadOfDepartment) VALUES
(1, 'Computer Science', 'Dr. Yann LeCun'), (2, 'Mathematics', 'Dr. Terence Tao'),
(3, 'Physics', 'Dr. Michio Kaku'), (4, 'Biology', 'Dr. Jennifer Doudna'),
(5, 'History', 'Dr. Yuval Noah Harari'), (6, 'Literature', 'Dr. Toni Morrison'),
(7, 'Chemistry', 'Dr. Carolyn Bertozzi'), (8, 'Engineering', 'Dr. Gwynne Shotwell'),
(9, 'Economics', 'Dr. Esther Duflo'), (10, 'Philosophy', 'Dr. Judith Butler');

-- Populating Faculty (Designations & Staff)
INSERT INTO Faculty (FacultyID, Name, Designation, DepartmentID) VALUES
(101, 'Alex Rivera', 'Senior Professor', 1), (102, 'Jordan Lee', 'Researcher', 2),
(103, 'Casey Morgan', 'Lead Lecturer', 3), (104, 'Taylor Swift', 'Visiting Professor', 4),
(105, 'Morgan Freeman', 'Professor Emeritus', 5), (106, 'Skyler White', 'Associate Professor', 6),
```

```
SELECT TOP (1000) [CourseID]
, [CourseName]
, [Credits]
, [DepartmentID]
, [FacultyID]
FROM [UniversityManagement].[dbo].[Courses]
```

CourseID	CourseName	Credits	DepartmentID	FacultyID
501	Machine Learning	4	1	101
502	Cybersecurity	4	1	101
503	Abstract Algebra	3	2	102
504	Number Theory	3	2	102
505	Astrophysics	4	3	103
506	Particle Physics	3	3	103
507	CRISPR Technology	4	4	104
508	Marine Biology	4	4	104
509	Digital History	3	5	105
510	Post-Colonialism	3	5	105

UniversityManagement

Object Explorer

- Database Diagrams
- Tables
 - System Tables
 - FileTables
 - External Tables
 - Graph Tables
 - dbo.Courses
 - dbo.Departments
 - Columns
 - Keys
 - Constraints
 - Triggers
 - Indexes
 - Statistics
 - dbo.Enrollments
 - dbo.Faculty
 - dbo.Library
 - dbo.Students
 - Dropped Ledger Tables
- Views
- External Resources
- Synonyms
- Programmability
- Query Store
- Service Broker

SQLQuery20...A\User (61)

```
1 SELECT TOP (1000) [DepartmentID]
2      ,[DepartmentName]
3      ,[HeadOfDepartment]
4      FROM [UniversityManagement].[dbo].[Departments]
5
```

89 % No issues found Ln: 1, Ch: 1 SPC CRLF Windows 1252

Results Messages

DepartmentID	DepartmentName	HeadOfDepartment
1	Computer Science	Dr. Yann LeCun
2	Mathematics	Dr. Terence Tao
3	Physics	Dr. Michio Kaku
4	Biology	Dr. Jennifer Doudna
5	History	Dr. Yuval Noah Harari
6	Literature	Dr. Toni Morrison
7	Chemistry	Dr. Carolyn Bertozzi
8	Engineering	Dr. Gwynne Shotwell
9	Economics	Dr. Esther Duflo
10	Philosophy	Dr. Judith Butler

UniversityManagement

Object Explorer

- Database Diagrams
- Tables
 - System Tables
 - FileTables
 - External Tables
 - Graph Tables
 - dbo.Courses
 - dbo.Departments
 - dbo.Enrollments
 - Columns
 - EnrollmentID (PK, int, not null)
 - StudentID (FK, int, not null)
 - CourseID (FK, int, not null)
 - Grade (decimal(3,2), null)
 - Keys
 - Constraints
 - Triggers
 - Indexes
 - Statistics
 - dbo.Faculty
 - dbo.Library
 - dbo.Students
 - Dropped Ledger Tables
- Views
- External Resources
- Synonyms
- Programmability
- Query Store
- Service Broker

SQLQuery21...A\User (106)

```
1 SELECT TOP (1000) [EnrollmentID]
2      ,[StudentID]
3      ,[CourseID]
4      ,[Grade]
5      FROM [UniversityManagement].[dbo].[Enrollments]
6
```

89 % No issues found Ln: 1, Ch: 1 SPC CRLF Windows 1252

Results Messages

EnrollmentID	StudentID	CourseID	Grade
1	1	501	4.00
2	1	502	3.70
3	2	501	3.90
4	3	503	3.80
5	4	504	3.50
6	5	505	3.90
7	6	506	3.40
8	7	507	4.00
9	9	509	3.20
10	10	510	3.10

UniversityManagement

Object Explorer

- Database Diagrams
- Tables
 - System Tables
 - FileTables
 - External Tables
 - Graph Tables
 - dbo.Courses
 - dbo.Departments
 - dbo.Enrollments
 - dbo.Faculty
 - Columns
 - FacultyID (PK, int, not null)
 - Name (varchar(100), not null)
 - Designation (varchar(50), null)
 - DepartmentID (FK, int, null)
 - Keys
 - Constraints
 - Triggers
 - Indexes
 - Statistics
 - dbo.Library
 - dbo.Students
 - Dropped Ledger Tables
- Views
- External Resources
- Synonyms
- Programmability
- Query Store
- Service Broker

SQLQuery22...A\User (82)

```
1 SELECT TOP (1000) [FacultyID]
2      ,[Name]
3      ,[Designation]
4      ,[DepartmentID]
5      FROM [UniversityManagement].[dbo].[Faculty]
6
```

89 % No issues found Ln: 1, Ch: 1 SPC CRLF Windows 1252

Results Messages

FacultyID	Name	Designation	DepartmentID
101	Alex Rivera	Senior Professor	1
102	Jordan Lee	Researcher	2
103	Casey Morgan	Lead Lecturer	3
104	Taylor Swift	Visiting Professor	4
105	Morgan Freeman	Professor Emeritus	5
106	Skyler White	Associate Professor	6
107	Jesse Pinkman	Lab Director	7
108	Riley Reid	Senior Engineer	8
109	Quinn Fabray	Lecturer	9
110	Avery Brooks	Dean of Philosophy	10

The screenshot shows the SQL Server Enterprise Manager interface. On the left, the Object Explorer displays the database structure for 'UniversityManagement', including tables like 'dbo.Students', 'dbo.Courses', and 'dbo.Faculty'. The main window shows a SQL query window with the following query:

```
1 SELECT TOP (1000) [StudentID]
2     , [Name]
3     , [DOB]
4     , [DepartmentID]
5 FROM [UniversityManagement].[dbo].[Students]
6
```

The query results are displayed in a table with the following data:

StudentID	Name	DOB	DepartmentID
1	Zoe Miller	2005-06-15	1
2	Liam Chen	2004-10-10	1
3	Sophia Khan	2006-02-22	2
4	Noah Garcia	2005-12-01	2
5	Emma Stone	2004-04-18	3
6	Lucas Grey	2005-08-25	3
7	Mia Wong	2006-01-05	4
8	Ethan Hunt	2005-09-12	4
9	Olivia Pope	2004-11-30	5
10	Jack Sparrow	2005-03-14	5

The status bar at the bottom indicates 'Query executed successfully.' and '10 rows'.

Part 4: Data Manipulation

1. Basic CRUD Operations:

a. Write SQL queries to:

- Add a new student, course, and faculty member.
- Update a student's department and a course's credits.
- Delete a faculty member.

```
118
119
120  --Data manipulation
121  -- Add new records
122  INSERT INTO Students (StudentID, Name, DOB, DepartmentID) VALUES (11, 'Kevin Hart', '2002-04-10', 1);
123  INSERT INTO Courses (CourseID, CourseName, Credits, DepartmentID, FacultyID) VALUES (511, 'AI Ethics', 3, 1, 101);
124  INSERT INTO Faculty (FacultyID, Name, Designation, DepartmentID) VALUES (111, 'Dr. Strange', 'Adjunct', 3);
125
126
127
```

89 % 4 0 89 % Ln: 120, Ch: 20 SPC LF Windows 1252

Messages

(1 row affected)

(1 row affected)

(1 row affected)

Completion time: 2026-02-09T12:06:35.9448382-05:00

```
126
127
128  -- Update operations
129  UPDATE Students SET DepartmentID = 2 WHERE StudentID = 1;
130  UPDATE Courses SET Credits = 5 WHERE CourseID = 501;
131
132
133
134
135
```

89 % 4 0 89 % Ln: 133, Ch: 1 SPC LF Windows

Messages

(1 row affected)

(1 row affected)

Completion time: 2026-02-09T12:08:57.9671540-05:00

```
130
131
132  -- Delete operations
133  DELETE FROM Faculty WHERE FacultyID = 111;
134
135
136
137
```

89 % 4 0 89 % Ln: 133, Ch: 1 SPC LF Windows 1

Messages

(1 row affected)

Completion time: 2026-02-09T12:09:58.2799327-05:00

Complex Queries:

- Retrieve:
 - The list of all students enrolled in a specific course.
 - The department-wise count of students.
 - The details of books borrowed by students in a specific department.

133 --Complex queries
 134
 135
 136 -- 1. List all students enrolled in 'Data Structures'
 137 SELECT s.Name
 138 FROM Students s
 139 JOIN Enrollments e ON s.StudentID = e.StudentID
 140 JOIN Courses c ON e.CourseID = c.CourseID
 141 WHERE c.CourseName = 'Data Structures';

89 % 3 0 1 2 Ln: 139, Ch: 48 SPC

Results Messages

Name

144 -- 2. Department-wise count of students
 145 SELECT d.DepartmentName, COUNT(s.StudentID) as StudentCount
 146 FROM Departments d
 147 LEFT JOIN Students s ON d.DepartmentID = s.DepartmentID
 148 GROUP BY d.DepartmentName;
 149
 150
 151
 152
 153
 154
 155
 156
 157

89 % 3 0 1 2 Ln: 155, Ch: 1

Results Messages

DepartmentName	StudentCount
Computer Science	2
Mathematics	3
Physics	2
Biology	2
History	2
Literature	0
Chemistry	0
Engineering	0
Economics	0
Philosophy	0

150 -- 3. Books borrowed by students in 'Computer Science'
 151 SELECT l.Title, s.Name
 152 FROM Library l
 153 JOIN Students s ON l.BorrowerID = s.StudentID
 154 JOIN Departments d ON s.DepartmentID = d.DepartmentID
 155 WHERE d.DepartmentName = 'Computer Science';
 156
 157

89 % 3 0 1 2 Ln: 154, Ch: 46 SPC LF Windows 1252

Results Messages

Title	Name
Sapiens	Liam Chen

Find:

- The top 3 students with the highest GPA.
- The faculty members teaching the most courses.

```
156
157 -- 4. Top 3 students with highest GPA (average grade)
158 SELECT TOP 3 s.Name, AVG(e.Grade) as GPA
159 FROM Students s
160 JOIN Enrollments e ON s.StudentID = e.StudentID
161 GROUP BY s.Name
162 ORDER BY GPA DESC;
163
```

89 % 3 0 ↑ ↓ Ln: 162, Ch: 19 SPC

Results Messages

	Name	GPA
1	Mia Wong	4.000000
2	Liam Chen	3.900000
3	Emma Stone	3.900000

```
164
165 -- 5. Faculty teaching the most courses
166 SELECT TOP 1 f.Name, COUNT(c.CourseID) as CourseCount
167 FROM Faculty f
168 JOIN Courses c ON f.FacultyID = c.FacultyID
169 GROUP BY f.Name
170 ORDER BY CourseCount DESC;
171
```

89 % 3 0 ↑ ↓ Ln: 169, Ch: 27 SPC LF Windows

Results Messages

	Name	CourseCount
1	Alex Rivera	3

Part 5: Stored Procedures

1. Stored Procedures:
 - a. Write a stored procedure to add a new student and enroll them in multiple courses.
 - b. Write a stored procedure to calculate the average grade for a course.

```
173 --STORED PROCEDURE
174
175 -- Procedure: Add student and enroll in a course
176 CREATE PROCEDURE sp_AddStudentAndEnroll
177     @ID INT, @Name VARCHAR(100), @DOB DATE, @DeptID INT, @CourseID INT
178 AS
179 BEGIN
180     BEGIN TRANSACTION;
181     INSERT INTO Students (StudentID, Name, DOB, DepartmentID) VALUES (@ID, @Name, @DOB, @DeptID);
182     INSERT INTO Enrollments (EnrollmentID, StudentID, CourseID, Grade) VALUES ((SELECT ISNULL(MAX(EnrollmentID), 0) + 1), @ID, @CourseID, 0);
183     COMMIT TRANSACTION;
184 END;
185 GO
186
187 -- Procedure: Calculate Average Grade for a Course
188
189 -- Procedure: Calculate Average Grade for a Course
190 CREATE PROCEDURE sp_GetAvgCourseGrade
191     @CourseID INT
192 AS
193 BEGIN
194     SELECT AVG(Grade) FROM Enrollments WHERE CourseID = @CourseID;
195 END;
196 GO
197
198
```

89 % 3 0 | ↑ ↓ Ln: 185, Ch: 3 | SPC | LF | Windows 1252

Messages

Commands completed successfully.

Completion time: 2026-02-09T12:17:37.2591931-05:00

89 % 3 0 | ↑ ↓ Ln: 194, Ch: 3 | SPC | LF | Windows 1252

Messages

Commands completed successfully.

Completion time: 2026-02-09T12:18:34.8459800-05:00

Part 6: Query Optimization

1. SARGable Queries:
 - a. Rewrite non-SARGable queries to ensure optimal index usage.
 - b. Example: Filter students based on GPA or department using indexed columns.

Part 7: Concurrency and Transactions

1. Simulate Concurrency:
 - a. Create a scenario where two students try to borrow the same book at the same time.
 - b. Implement transaction isolation levels to prevent concurrency issues.
Hint: create a insert statement for Library and wrap it in a TRANSACTION statement
2. Transactional Integrity:
 - a. Ensure that a course cannot be deleted if students are enrolled in it.

b. Implement cascading actions (e.g., ON DELETE SET NULL for Library table borrow records).

```
197 --Concurrency and transaction
198 -- Simulate borrowing with Serialized Isolation to prevent double-booking
199 SET TRANSACTION ISOLATION LEVEL SERIALIZABLE;
200 BEGIN TRANSACTION;
201 -- Check if book is available (BorrowerID is NULL)
202 IF EXISTS (SELECT 1 FROM Library WHERE BookID = 1001 AND BorrowerID IS NULL)
203 BEGIN
204     UPDATE Library SET BorrowerID = 1, DueDate = '2026-03-01' WHERE BookID = 1001;
205     PRINT 'Book borrowed successfully';
206 END
207 ELSE
208 BEGIN
209     PRINT 'Book already borrowed';
210 END
211 COMMIT TRANSACTION;
```

89 % 1 0 89 % Ln: 211, Ch: 20 SPC LF Windows 1252

Messages

Book already borrowed

Completion time: 2026-02-09T12:19:38.0925392-05:00

```
212
213 -- Transactional Integrity: Prevent deleting course if students are enrolled
214 -- This is handled by Foreign Key constraints. To implement cascading:
215 ALTER TABLE Library
216 DROP CONSTRAINT [FK__Library__Borrowe__5CD6CB2B]; -- Replace with actual constraint name
217
218
219
220
221
```

89 % 1 0 89 % Ln: 223, Ch: 1 SPC LF Windows 1252

Messages

Commands completed successfully.

Completion time: 2026-02-09T12:24:13.8846407-05:00

```
219
220 ALTER TABLE Library
221 ADD CONSTRAINT FK_Library_Student
222 FOREIGN KEY (BorrowerID) REFERENCES Students(StudentID)
223 ON DELETE SET NULL; -- If a student is deleted, their borrow record stays but becomes anonymous.
224
225
226
227
228
```

89 % 1 0 89 % Ln: 224, Ch: 1 SPC LF Windows 125

Messages

Commands completed successfully.

Completion time: 2026-02-09T12:25:05.4405987-05:00

Generate Reports:

- Write queries to generate:
 - A list of students with overdue books.

- A summary report of department-wise course enrollments.
- A ranking of students by GPA, grouped by department.

224
225 -- 1. List of students with overdue books (assuming current date is Feb 9, 2026)
226 SELECT s.Name, l.Title, l.DueDate
227 FROM Students s
228 JOIN Library l ON s.StudentID = l.BorrowerID
229 WHERE l.DueDate < '2026-02-09';
230
231
232
233
234

89 % 1 0 Ln: 229, Ch: 32 (207 chars, 5 lines) SPC LF Windows 1252

Results Messages

Name	Title	DueDate
------	-------	---------

235
236 -- 2. Summary report of department-wise course enrollments
237 SELECT d.DepartmentName, COUNT(e.EnrollmentID) AS TotalEnrollments
238 FROM Departments d
239 JOIN Courses c ON d.DepartmentID = c.DepartmentID
240 JOIN Enrollments e ON c.CourseID = e.CourseID
241 GROUP BY d.DepartmentName;
242
243
244

39 % 1 0 Ln: 243, Ch: 1 SPC LF Windows

Results Messages

	DepartmentName	TotalEnrollments
1	Biology	1
2	Computer Science	3
3	History	2
4	Mathematics	2
5	Physics	2

244
245 -- 3. Ranking students by GPA, grouped by department
246 SELECT d.DepartmentName, s.Name, AVG(e.Grade) AS GPA
247 FROM Departments d
248 JOIN Students s ON d.DepartmentID = s.DepartmentID
249 JOIN Enrollments e ON s.StudentID = e.StudentID
250 GROUP BY d.DepartmentName, s.Name
251 ORDER BY d.DepartmentName, GPA DESC;

89 % 1 0 Ln: 251, Ch: 37 SPC LF Windows 1252

Results Messages

	DepartmentName	Name	GPA
1	Biology	Mia Wong	4.000000
2	Computer Science	Liam Chen	3.900000
3	History	Olivia Pope	3.200000
4	History	Jack Sparrow	3.100000
5	Mathematics	Zoe Miller	3.850000
6	Mathematics	Sophia Khan	3.800000
7	Mathematics	Noah Garcia	3.500000
8	Physics	Emma Stone	3.900000
9	Physics	Lucas Grey	3.400000

Query executed successfully. ENIOLA (17.0 RTM) ENIOLA\User (55) UniversityManagement 00:00:00 Row: 1, Col: 1 9 rows

UniversityManagement

Object Explorer

- Database Diagrams
- Tables
 - System Tables
 - FileTables
 - External Tables
 - Graph Tables
 - dbo.Courses
 - dbo.Departments
 - dbo.Enrollments
 - dbo.Faculty
 - dbo.Library
 - dbo.Students
 - Columns
 - StudentID (PK, int, not null)
 - Name (varchar(100), not null)
 - DOB (date, null)
 - DepartmentID (FK, int, null)
 - Keys
 - Constraints
 - Triggers
 - Indexes
 - Statistics
 - Dropped Ledger Tables
- Views
- External Resources
- Synonyms

SQLQuery25...A\User (85)

```

1 SELECT TOP (1000) [CourseID]
2     , [CourseName]
3     , [Credits]
4     , [DepartmentID]
5     , [FacultyID]
6 FROM [UniversityManagement].[dbo].[Courses]
7

```

89 % No issues found Ln: 1, Ch: 1 SPC CRLF Windows 1252

Results Messages

	CourseID	CourseName	Credits	DepartmentID	FacultyID
1	501	Machine Learning	5	1	101
2	502	Cybersecurity	4	1	101
3	503	Abstract Algebra	3	2	102
4	504	Number Theory	3	2	102
5	505	Astrophysics	4	3	103
6	506	Particle Physics	3	3	103
7	507	CRISPR Technology	4	4	104
8	508	Marine Biology	4	4	104
9	509	Digital History	3	5	105
10	510	Post-Colonialism	3	5	105
11	511	AI Ethics	3	1	101

Query executed successfully. ENIOLA (17.0 RTM) ENIOLA\User (85) UniversityManagement 00:00:00 Row: 1, Col: 1 11 rows

UniversityManagement

Object Explorer

- Database Diagrams
- Tables
 - System Tables
 - FileTables
 - External Tables
 - Graph Tables
 - dbo.Courses
 - dbo.Departments
 - dbo.Enrollments
 - dbo.Faculty
 - dbo.Library
 - dbo.Students
 - Columns
 - StudentID (PK, int, not null)
 - Name (varchar(100), not null)
 - DOB (date, null)
 - DepartmentID (FK, int, null)
 - Keys
 - Constraints
 - Triggers
 - Indexes
 - Statistics
 - Dropped Ledger Tables
- Views
- External Resources
- Synonyms

SQLQuery26...A\User (87)

```

1 SELECT TOP (1000) [DepartmentID]
2     , [DepartmentName]
3     , [HeadOfDepartment]
4 FROM [UniversityManagement].[dbo].[Departments]
5

```

89 % No issues found Ln: 1, Ch: 1 SPC CRLF Windows 1252

Results Messages

	DepartmentID	DepartmentName	HeadOfDepartment
1	1	Computer Science	Dr. Yann LeCun
2	2	Mathematics	Dr. Terence Tao
3	3	Physics	Dr. Michio Kaku
4	4	Biology	Dr. Jennifer Doudna
5	5	History	Dr. Yuval Noah Harari
6	6	Literature	Dr. Toni Morrison
7	7	Chemistry	Dr. Carolyn Bertozzi
8	8	Engineering	Dr. Gwynne Shotwell
9	9	Economics	Dr. Esther Duflo
10	10	Philosophy	Dr. Judith Butler

Query executed successfully. ENIOLA (17.0 RTM) ENIOLA\User (87) UniversityManagement 00:00:00 Row: 1, Col: 1 10 rows

UniversityManagement

Object Explorer

- Database Diagrams
- Tables
 - System Tables
 - FileTables
 - External Tables
 - Graph Tables
 - dbo.Courses
 - dbo.Departments
 - dbo.Enrollments
 - dbo.Faculty
 - dbo.Library
 - dbo.Students
 - Columns
 - StudentID (PK, int, not null)
 - Name (varchar(100), not null)
 - DOB (date, null)
 - DepartmentID (FK, int, null)
 - Keys
 - Constraints
 - Triggers
 - Indexes
 - Statistics
 - Dropped Ledger Tables
- Views
- External Resources
- Synonyms

SQLQuery27...A\User (89)

```

1 SELECT TOP (1000) [EnrollmentID]
2     , [StudentID]
3     , [CourseID]
4     , [Grade]
5 FROM [UniversityManagement].[dbo].[Enrollments]
6

```

89 % No issues found Ln: 1, Ch: 1 SPC CRLF Windows 1252

Results Messages

	EnrollmentID	StudentID	CourseID	Grade
1	1	1	501	4.00
2	2	1	502	3.70
3	3	2	501	3.90
4	4	3	503	3.80
5	5	4	504	3.50
6	6	5	505	3.90
7	7	6	506	3.40
8	8	7	507	4.00
9	9	9	509	3.20
10	10	10	510	3.10

Query executed successfully. ENIOLA (17.0 RTM) ENIOLA\User (89) UniversityManagement 00:00:00 Row: 1, Col: 1 10 rows

UniversityManagement

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 - dbo.Faculty
 - dbo.Library
 - dbo.Students
 - Columns
 - StudentID (PK, int, not null)
 - Name (varchar(100), not null)
 - DOB (date, null)
 - DepartmentID (FK, int, null)
 - Keys
 - Constraints
 - Triggers
 - Indexes
 - Statistics
 - Dropped Ledger Tables
- Views
- External Resources
- Synonyms

SQLQuery28...User (97)*

```
1 SELECT TOP (1000) [FacultyID]
2     ,[Name]
3     ,[Designation]
4     ,[DepartmentID]
5 FROM [UniversityManagement].[dbo].[Faculty]
6
```

89 % No issues found Ln: 1, Ch: 1 SPC CRLF Windows 1252

Results Messages

	FacultyID	Name	Designation	DepartmentID
1	101	Alex Rivera	Senior Professor	1
2	102	Jordan Lee	Researcher	2
3	103	Casey Morgan	Lead Lecturer	3
4	104	Taylor Swift	Visiting Professor	4
5	105	Morgan Freeman	Professor Emeritus	5
6	106	Skyler White	Associate Professor	6
7	107	Jesse Pinkman	Lab Director	7
8	108	Riley Reid	Senior Engineer	8
9	109	Quinn Fabray	Lecturer	9
10	110	Avery Brooks	Dean of Philosophy	10

Query executed successfully. ENIOLA (17.0 RTM) ENIOLA\User (97) UniversityManagement 00:00:00 Row: 1, Col: 1 10 rows

UniversityManagement

Object Explorer

- Database Diagrams
- Tables
 - System Tables
 - FileTables
 - External Tables
 - Graph Tables
 - dbo.Courses
 - dbo.Departments
 - dbo.Enrollments
 - dbo.Faculty
 - dbo.Library
 - dbo.Students
 - Columns
 - StudentID (PK, int, not null)
 - Name (varchar(100), not null)
 - DOB (date, null)
 - DepartmentID (FK, int, null)
 - Keys
 - Constraints
 - Triggers
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 - Statistics
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- Views
- External Resources
- Synonyms

SQLQuery29...A\User (99)

```
1 SELECT TOP (1000) [BookID]
2     ,[Title]
3     ,[Author]
4     ,[BorrowerID]
5     ,[DueDate]
6 FROM [UniversityManagement].[dbo].[Library]
7
```

89 % No issues found Ln: 1, Ch: 1 SPC CRLF Windows 1252

Results Messages

	BookID	Title	Author	BorrowerID	DueDate
1	1001	Deep Learning	Ian Goodfellow	1	2026-04-01
2	1002	Sapiens	Yuval Harari	2	2026-04-05
3	1003	A Brief History of Time	Hawking	3	2026-03-15
4	1004	The Gene	Mukherjee	5	2026-03-20
5	1005	Atomic Habits	James Clear	7	2026-04-10
6	1006	Thinking, Fast and Slow	Kahneman	9	2026-04-12
7	1007	Rust Programming	Klabnik	1	2026-04-15
8	1008	The Selfish Gene	Dawkins	8	2026-04-20
9	1009	Capital in the 21st Century	Piketty	4	2026-04-22
10	1010	Meditations	Marcus Aurelius	10	2026-04-25

Query executed successfully. ENIOLA (17.0 RTM) ENIOLA\User (99) UniversityManagement 00:00:00 Row: 1, Col: 1 10 rows

UniversityManagement

Object Explorer

- Database Diagrams
- Tables
 - System Tables
 - FileTables
 - External Tables
 - Graph Tables
 - dbo.Courses
 - dbo.Departments
 - dbo.Enrollments
 - dbo.Faculty
 - dbo.Library
 - dbo.Students
 - Columns
 - StudentID (PK, int, not null)
 - Name (varchar(100), not null)
 - DOB (date, null)
 - DepartmentID (FK, int, null)
 - Keys
 - Constraints
 - Triggers
 - Indexes
 - Statistics
 - Dropped Ledger Tables
- Views
- External Resources
- Synonyms

SQLQuery30...User (102)

```
1 SELECT TOP (1000) [StudentID]
2     ,[Name]
3     ,[DOB]
4     ,[DepartmentID]
5 FROM [UniversityManagement].[dbo].[Students]
6
```

89 % No issues found Ln: 1, Ch: 1 SPC CRLF Windows 1252

Results Messages

	StudentID	Name	DOB	DepartmentID
1	1	Zoe Miller	2005-06-15	2
2	2	Liam Chen	2004-10-10	1
3	3	Sophia Khan	2006-02-22	2
4	4	Noah Garcia	2005-12-01	2
5	5	Emma Stone	2004-04-18	3
6	6	Lucas Grey	2005-08-25	3
7	7	Mia Wong	2006-01-05	4
8	8	Ethan Hunt	2005-09-12	4
9	9	Olivia Pope	2004-11-30	5
10	10	Jack Sparrow	2005-03-14	5
11	11	Kevin Hart	2002-04-10	1

Query executed successfully. ENIOLA (17.0 RTM) ENIOLA\User (102) UniversityManagement 00:00:00 Row: 1, Col: 1 11 rows

Documentation Report

Database Design

The database utilizes a relational schema to manage university operations. It connects six primary entities: Departments, Faculty, Students, Courses, Enrollments, and Library records. Relationships are established via Foreign Keys (e.g., DepartmentID linking Students to Departments).

Normalization Process

The design adheres to Third Normal Form (3NF):

1NF: Each table has a Primary Key and contains atomic values.

2NF: All non-key attributes are fully dependent on the Primary Key.

3NF: Eliminated transitive dependencies (e.g., Faculty details are in a separate table from Courses).

Query Optimization Techniques

To ensure performance, the following techniques were used:

Indexing: Applying Primary Keys naturally creates indexes for faster JOIN operations.

Specific Selection: Using SELECT ColumnNames instead of SELECT * to reduce data overhead.

Efficient Joins: Utilizing INNER JOIN for mandatory relationships and LEFT JOIN only when counting entities that might be zero.